11 May 2020
Anirban & Margaret

Summary of the prospective thermospore biogeography manuscript by Margaret and Anirban

Title: Spatial biogeography of thermospores in North Atlantic marine sediment

Summary: This study will use thermospore ASVs from Margaret and Emily Ellefson's thermospore incubation experiments to assess the distribution of thermospores within marine sediments of the North Atlantic and Canadian Arctic Archipelago (Figure 1). Comparisons of thermospore communities in different geographic regions will allow the identification of endemic and cosmopolitan ASVs, leading to inferences on the influence of different dispersal vectors (e.g. water currents). Possible dispersal vectors include the North Atlantic Subpolar Gyre, Baffin Island Current, and Greenland Current (Figure 2). We may include predictive discussions about possible high-temperature source environments of these thermospores, including the mid-ocean ridge spreading centres (e.g. Mid-Atlantic Ridge, Gakkal Ridge), oil and gas seeps (e.g. Scott Inlet), and oil production facilities of Northern nations. We will integrate any such discussion in an open-ended manner so that the main focus of the text remains on the biogeography and water currents instead of trying to trace the source. This study will incorporate findings from Margaret's master's thesis. Chapter 4: *Lateral biogeography of thermophilic endospores in North Atlantic sediment.*

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Figure 1: Locations of the sediment used in thermospore incubations for this study. Yellow stars identify locations used in Margaret's thesis. Red stars identify locations of sediment used in Emily's thermospore incubations.

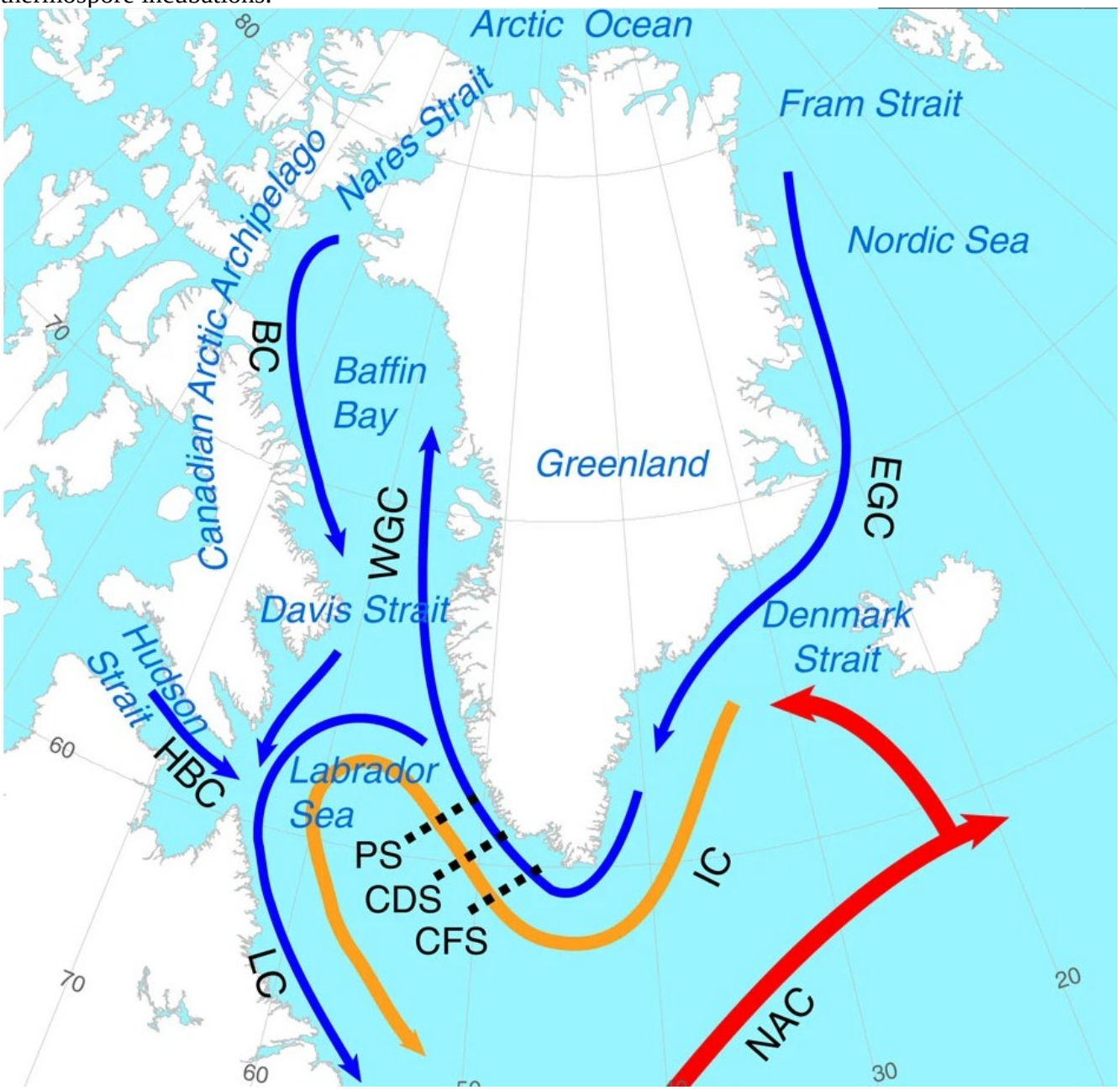


Figure 2: Currents of the western North Atlantic.
*Figure from: Yang, Q., Dixon, T., Myers, P. et al. Recent increases in Arctic freshwater flux affects Labrador Sea convection and Atlantic overturning circulation. Nat Commun* ***7,*** *10525 (2016). https://doi.org/10.1038/ncomms10525*